

Marco Balduini

Nationality: Italian

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PROFILE

Co-founder and CEO of Quantia Consulting and co-founder and COO of Motus ml. PhD from Politecnico di Milano, with more than 15 years of experience in semantic data integration, Knowledge Graphs, stream processing and enterprise AI. Leads AI and data engineering projects for industrial clients in the aerospace, defence and semiconductor sectors, coordinates technical work packages in Horizon Europe and regionally funded R&D projects, and designs and delivers Data & AI training programmes in partnership with Politecnico di Milano.

WORK EXPERIENCE

01/2023 - CURRENT · MILAN, ITALY

Co-founder and COO – Motus ml S.r.l. S.B.

Cloud-independent AI, edge and real-time analytics.

- Leads the delivery of AI and data engineering projects for major aerospace and defence manufacturers, from discovery and scoping to deployment at client sites.
- Designs hybrid architectures in which LLMs support semantic understanding and reconciliation while deterministic, auditable logic produces the final results; on-premise and air-gapped deployments with local LLM inference.
- Delivered solutions include a RAG-based assistant for engineering documentation, a streaming data platform with rule-based validation, a Knowledge Graph-based semantic data backbone and a Virtual Knowledge Graph supporting the qualification of mechanical processes.

2019 - CURRENT · VARESE, ITALY

Co-founder and CEO – Quantia Consulting S.r.l.

Data & AI education and strategic consultancy.

- Sole director of the company, which operates in two business lines: Data & AI training and funded R&D and innovation projects.
- Designs the Data & AI training catalogue, from data culture to AI for experts (including Generative AI, RAG and Agentic AI), and delivers corporate programmes.
- Leads data platform assessments and target architecture definition for industrial clients, e.g. AS-IS assessment and TO-BE guidelines for the high-volume measurement data repository of a semiconductor manufacturer.
- Technical and operational lead for Quantia in Horizon Europe and Regione Lombardia funded projects (see Projects).

09/2015 - 10/2020 · COMO, ITALY

Co-founder and Partner – Fluxedo S.r.l.

Real-time social media analysis, crowd analytics and app development.

11/2011 - 10/2014 · MILAN, ITALY

Research Assistant – Politecnico di Milano, Department of Electronics, Information and Bioengineering (DEIB)

Contributed to a general infrastructure merging semantic streams, social media and sensor networks, within the international projects EIT ICT Labs City Data Fusion for Event Management (14189) and MODAClouds (FP7-318484).

06/2011 - 09/2011 · MILAN, ITALY

Research Assistant – Cefriel S.Cons.r.l.

Contributed to the European project LarkC (IST-FP7-2007-1stCall), developing an integrated environment to manage social media data collected by the Korean company Saltlux and analyse it in real time with the C-SPARQL engine.

TEACHING & TRAINING

2019 - CURRENT · ITALY

Data & AI Educator and Programme Designer

Over 100 training engagements delivered for universities, corporate clients and public administrations, from half-day workshops to multi-module curricula, in Italian and English, in person and online.

- **Post-graduate education:** programme direction, curriculum design and teaching in 3 post-graduate master's and continuing-education programmes on AI project management, data science and applied AI.
- **Technical courses:** data engineering, real-time data processing, semantic data integration and Knowledge Graphs, machine learning, Generative AI, RAG and agentic AI, edge AI, with hands-on labs on a dedicated JupyterHub-based platform.
- **Business courses:** data culture, data-driven decision making, AI awareness and adoption, AI project and product management.
- **Public sector:** courses for public administration staff on service design and requirements elicitation for public procurement.

PROJECTS

09/2026 - CURRENT · HORIZON EUROPE

TRIAD – Balancing Foundational and AI Competences for Inclusive, Well-being-Oriented Education

Technical and operational focal point for Quantia Consulting and leader of WP7 (Data Enrichment, Integration and Exploratory Dashboard): consolidation of pilot data and enrichment with external datasets, including SES indicators, OECD-TALIS and urban environmental metrics (T7.1); design and development of the interactive web dashboard (T7.2); definition of APIs, metadata and integration protocols compliant with FAIR and Frictionless Data standards (T7.3). Coordinates SME engagement and the alignment of AI skills with labour market needs (T8.3), and policy dialogues and round tables with ministries and relevant authorities (T10.3). Main point of contact for the coordinator (RPTU) and the partners, including finalisation of the Consortium Agreement and review of deliverables.

12/2025 - CURRENT · REGIONE LOMBARDIA, BANDO TECNOLOGIE STRATEGICHE

AI4SUM – AI-Safety 4 Smart Urban Mobility

Technical and operational focal point for Quantia Consulting on data management, integration and analysis, coordinating across the operational units: data catalogue of the sources required by the use cases, including extraction of sources that are not natively open; pre-processing strategies and harmonisation of geographic bases (e.g. the Milan road graph); risk mitigation strategy based on static and open data, ensuring the system's functional autonomy from the partners' real-time streams; technical coordination with the industrial partners and data quality verification; dissemination and administrative reporting.

06/2020 - 06/2025 · HORIZON 2020

Equal-Life – Early Environmental Quality and Life-Course Mental Health Effects

Research project on the early-life exposome, integrating physical, social and internal exposure data to study their impact on children's mental health and development. Technical and operational leader of WP9 (Tools, Training and Implementation): requirements definition from the outputs and data formats of the other work packages, and development of the Equal-Life Toolkit, a software and documentation toolkit enabling the reuse and replication of the project's exposome data integration and exploration approaches.

EDUCATION

2014 - 2019 · MILAN, ITALY

PhD in Information and Communication Technologies – Politecnico di Milano

Thesis: *On the continuous and reactive analysis of a variety of spatio-temporal data*

2008 - 2011 · MILAN, ITALY

Master's Degree in Information and Communication Technologies – Politecnico di Milano

Thesis: *SLD4SM: an integrated environment for managing social media streams through linked data* (original title in Italian)

LANGUAGE SKILLS

Mother tongue: **Italian**

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C2	C1	C2	C2

SELECTED PUBLICATIONS

LLM-Enhanced Semantic Data Integration of Electronic Component Qualifications in the Aerospace Domain (2026)A. De Santis, **M. Balduini**, M. Belcao, A. Proia, M. Brambilla, E. Della Valle. *Proc. of ESWC 2026*, LNCS 16550: 398–415.**Integrating Large Language Models and Knowledge Graphs for Extraction and Validation of Textual Test Data** (2024)A. De Santis, **M. Balduini**, F. De Santis, A. Proia, A. Leo, M. Brambilla, E. Della Valle. *Proc. of ISWC 2024*, Part III: 304–323.**On Teaching Web Stream Processing – Lessons Learned** (2020)R. Tommasini, E. Della Valle, **M. Balduini**, S. Sakr. *Proc. of ICWE 2020*: 485–493.**Models and Practices in Urban Data Science at Scale** (2019)**M. Balduini**, M. Brambilla, E. Della Valle, C. Marazzi, T. Arabghalizi, B. Rahdari, M. Vescovi. *Big Data Research* 17: 66–84.**Cost-Aware Streaming Data Analysis: Distributed vs Single-Thread** (2018)**M. Balduini**, S. Pasupathipillai, E. Della Valle. *Proc. of DEBS 2018*: 160–170.**VoCaLS: Vocabulary and Catalog of Linked Streams** (2018)R. Tommasini, Y. Abo Sedira, D. Dell'Aglio, **M. Balduini**, M. I. Ali, D. Le Phuoc, E. Della Valle, J.-P. Calbimonte. *Proc. of ISWC 2018*.**SLD Revolution: A Cheaper, Faster yet More Accurate Streaming Linked Data Framework** (2017)**M. Balduini**, E. Della Valle, R. Tommasini. *ESWC 2017 Satellite Events*: 263–279.**TripleWave: Spreading RDF Streams on the Web** (2016)A. Mauri, J.-P. Calbimonte, D. Dell'Aglio, **M. Balduini**, M. Brambilla, E. Della Valle, K. Aberer. *Proc. of ISWC 2016*, Part II: 140–149.**Heaven: A Framework for Systematic Comparative Research Approach for RSP Engines** (2016)R. Tommasini, E. Della Valle, **M. Balduini**, D. Dell'Aglio. *Proc. of ESWC 2016*: 250–265.**CitySensing: Fusing City Data for Visual Storytelling** (2015)**M. Balduini**, E. Della Valle, M. Azzi, R. Larcher, F. Antonelli, P. Ciuccarelli. *IEEE MultiMedia* 22(3): 44–53.**Reality Mining on Micropost Streams: Deductive and Inductive Reasoning for Personalized and Location-Based Recommendations** (2014)**M. Balduini**, I. Celino, D. Dell'Aglio, E. Della Valle, Y. Huang, T. Lee, S.-H. Kim, V. Tresp. *Semantic Web* 5(5): 341–356.**Social Listening of City Scale Events Using the Streaming Linked Data Framework** (2013)**M. Balduini**, E. Della Valle, D. Dell'Aglio, M. Tsytsarau, T. Palpanas, C. Confalonieri. *Proc. of ISWC 2013*, Part II: 1–16.**BOTTARI: An Augmented Reality Mobile Application to Deliver Personalized and Location-Based Recommendations by Continuous Analysis of Social Media Streams** (2012)**M. Balduini**, I. Celino, D. Dell'Aglio, E. Della Valle, Y. Huang, T. Lee, S.-H. Kim, V. Tresp. *Journal of Web Semantics* 16: 33–41.